

Can Agentic AI Settle a 20-Year-Old Open Problem?

The decidability of $\mathcal{ALCI}_{\text{RCC5}}$ — a machine-checked campaign

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The one-slide version

- In 2002/2003 I defined a family of **spatial description logics** and left their central decidability question **open**. It still is.
- In 2026 I directed a **four-month AI campaign** on it: two frontier models, adversarial review, machine verification.
- The question is **not settled**. But it is now *reduced* — honestly and partly kernel-checked — to **one well-posed lemma**, with a decidable fragment, a certified normal form, a working reasoner, and an exact arithmetical address (Π_1^0) along the way.

How to read everything in this talk

The mathematics was produced by AI assistants and is *unverified by human experts*.

Standing label: **strongly supported, not certified** — except the parts checked by the **Lean 4 kernel**, which are machine-verified (but still human-unreviewed).

Five ways two regions can relate (RCC5)



DR

disjoint



PO

partial overlap



EQ

equal



PP

proper part



PPI

proper superpart

- Every pair of regions stands in **exactly one** of these five relations.
- *Jointly exhaustive, pairwise disjoint* — the classic qualitative spatial calculus (a coarser cousin of RCC8).
- Think GIS, layout, biology, any “what contains / touches what” domain — **no coordinates required**.

Reasoning without geometry: the composition table

If I know $x R y$ and $y S z$, what can $x ? z$ be?

known	known	forced / allowed
$x PP y$	$y PP z$	$x PP z$ (<i>forced</i> : parts of parts are parts)
$x PP y$	$y DR z$	$x DR z$ (<i>forced</i> : inside something disjoint)
$x DR y$	$y DR z$	anything at all (<i>free</i> : 5 options)

- All 25 cells form the **composition table** — pure algebra, derived once from set semantics, then used forever.
- **Abstract semantics**: a “model” is any relation assignment closed under this table — *space itself has left the building*.
- Every probe in this project re-derives the table from scratch, so the algebra can never silently drift.

A description logic: you write definitions and facts

A **TBox** is a set of concept *definitions*, built from properties and **quantified spatial relations**:

$\text{CityAtRiver} \equiv \text{City} \sqcap \exists \text{PO}. \text{River}$ a city *overlapping* a river

$\text{GermanCity} \equiv \text{City} \sqcap \forall \text{PP}. (\neg \text{Country} \sqcup \text{Germany})$ a city in *no country but* Germany

Then we **assert** a few primitive facts — and nothing more:

$\text{Hamburg} \sqsubseteq \text{City} \sqcap \exists \text{PO}. \text{Alster}$ (a city, overlapping the Alster)

$\text{Alster} \sqsubseteq \text{River} \sqcap \exists \text{PP}. \text{Germany} \sqcap \forall \text{PO}. \neg \text{Country} \sqcap \forall \text{PP}. (\neg \text{Country} \sqcup \text{Germany})$

Note what we did *not* say: that Hamburg is a GermanCity, or a CityAtRiver.

... and the reasoner *infers* the classification

We never *asserted* that Hamburg is a GermanCity or a CityAtRiver — the **reasoner derives both**.

City at a river (immediate):

Hamburg PO Alster $\wedge$ Alster $\sqsubseteq$ River $\implies$ Hamburg $\sqsubseteq$ City $\sqcap \exists$ PO. River = CityAtRiver.

German city — needs the **composition table**. For any country c with Hamburg PP c :

- Alster PO Hamburg PP $c \Rightarrow$ Alster $\{PO, PP\} c$ comp(PO, PP) = $\{PO, PP\}$
- Alster overlaps no country ($\forall PO. \neg \text{Country}$) $\Rightarrow$ Alster PP c
- Alster's only country-superpart is Germany $\Rightarrow c = \text{Germany}$

So Hamburg *is* a German city — and every hidden fact reduces to **concept satisfiability**:

$C \sqsubseteq D$ iff $C \sqcap \neg D$ unsat.

$\mathcal{ALCI}_{\text{RCC5}}$: the spatial relations *become* the roles

The logic (Wessel 2002/2003):

- $\mathcal{ALCI}$ = the basic DL with inverse roles;
- roles = the five RCC5 relations themselves;
- the composition table is enforced **globally**;
- EQ = identity (*strong* semantics).

So one can say things like

$\forall \text{PP}. \neg \text{Lake} \quad \exists \text{DR}. \text{Reserve}$

“no part is a lake”, “disjoint from some reserve”.

Why this is different

Concrete-domain DLs constrain spatial *attributes* of objects. Here the relations are *first-class roles*: one can **quantify over them**. None of the known concrete-domain decidability results apply.

Why the standard toolbox dies on contact

Every pair of elements carries a relation $\Rightarrow$ a model is a **complete labelled graph**, never a tree.

no tree models

Blocking/unravelling —
the workhorse of DL
decidability — has no tree
to work on.

no finite models

$\exists P P I . T \sqcap \forall P P I . \exists P P I . T$
forces an infinite ascending
part-of tower.

nondeterministic table

Most cells allow several
outcomes — composition
constrains but rarely
determines.

Creating one new element forces relations to every existing element — and a dormant $\forall$
anywhere may wake up because of it.

The landscape: open exactly at the top

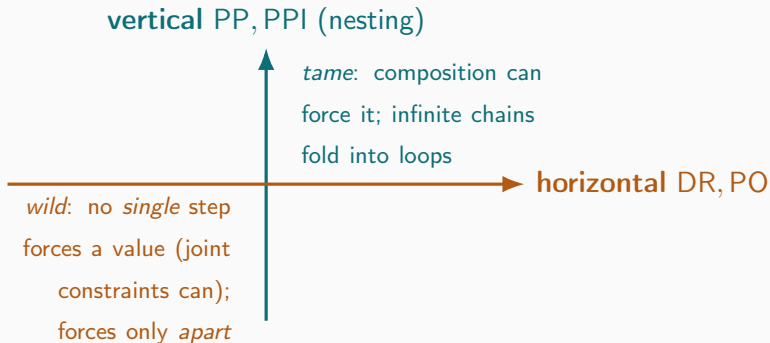
Logic	Relations	Concept satisfiability
$\mathcal{ALCI}_{\text{RCC1}}$	{SR}	decidable ($\equiv$ S5)
$\mathcal{ALCI}_{\text{RCC2}}$	{DR, O}	decidable
$\mathcal{ALCI}_{\text{RCC3}}$	{DR, ONE, EQ}	decidable (via two-variable FO!)
$\mathcal{ALCI}_{\text{RCC5}, \forall\text{PO-free}}$	RCC5 w/o $\forall\text{PO}$	decidable (this project)
$\mathcal{ALCI}_{\text{RCC5}}$	all five	open ; PSpace-hard
$\mathcal{ALCI}_{\text{RCC8}}$	eight RCC8 rel.	open ; ExpTime-hard

Lutz & Wolter 2006, after proving the *topological* cousins undecidable:

“Perhaps the most interesting candidate is $L_{\text{RCC5}}(\text{RS}) [= \mathcal{ALCI}_{\text{RCC5}}]$... to which the reduction exhibited in Section 8 does not apply.”

Recognized open by the leading experts — for **twenty years**.

One logic, two completely different axes



- Depth (nesting) is **never the problem** — ordinary modal-logic folding handles it.
- Everything hard lives in **horizontal crowds**: many pairwise-distinct neighbours that no finite summary seems to capture.

The asymmetry, made precise (machine-verified)

No *single* step forces a horizontal value. Of the 16 compositions of proper relations, exactly **four** return a single value:

$$\text{comp}(\text{DR}, \text{PPI}) = \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}, \quad \text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}, \quad \text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$$

— every one runs through PP/PPI; no single cell yields $\{\text{PO}\}$. But the *intersection of several can force PO* ($\text{comp}(\text{DR}, \text{PO}) \cap \text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}\}$) — so “shadow” is joint closure, not single-path.

Disjointness is half-deterministic.

downward: $x \text{ DR } y \Rightarrow$ every part of x is DR from every part of y

upward: a superpart of x may be DR, PO, or PPI to y — **free**

So one high DR edge dispatches an *arbitrarily wide* crowd of disjointnesses below it, for_{10/39} free. What resists: PO, and the upward direction.

Shadows, live width, and the one property that matters

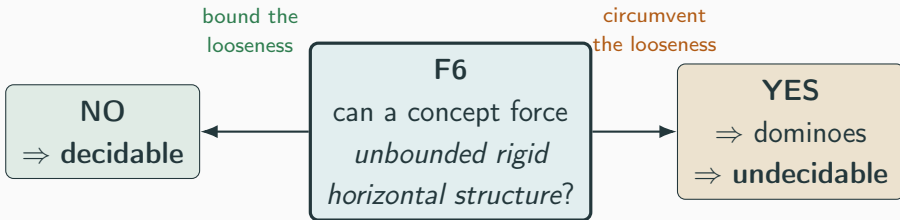
- An edge is a **shadow** if composition already forces it from edges you have — it costs a decision procedure *nothing*.
- The **live width** of an interface = the number of *non-shadow* edges there: the genuine branching that must be recorded.

Property F6 (the keystone of everything)

Is the live width of models bounded by a computable function of the input concept?

If yes $\Rightarrow$ finite certificate space $\Rightarrow$ **satisfiability is decidable**.

The knife's edge: one fact, two futures



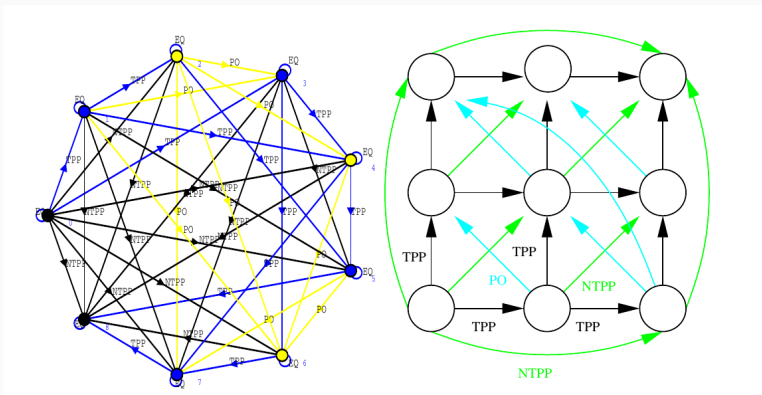
The *same* looseness of the composition table blocks **both** proofs. That is *why* neither direction has been settled in twenty years — it is one question seen from two sides.

The standard weapon needs a rigid grid

- Classic undecidability proofs encode **domino tilings** of $\mathbb{N} \times \mathbb{N}$: build a grid, force tile constraints, reduce the halting problem.
- The crux is **rigidity**: going *east then north* must land on *exactly the same cell* as *north then east* ($R_X \circ R_Y = R_Y \circ R_X$, the “coincidence condition”).
- Every known route into $\mathcal{ALCI}_{\text{RCC}}$ needs a feature it does not have: number restrictions, arbitrary role axioms, topological rigidity, nominals. . .

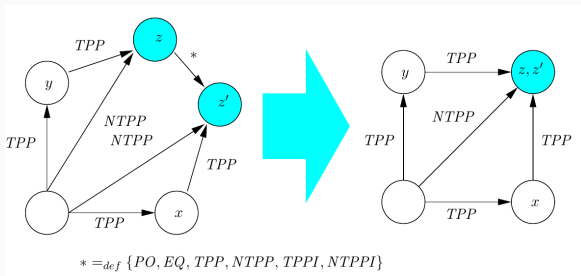
Question (2003 and today): can the composition table itself supply the rigidity?

2003: the grid models *exist*...



An RCC8 network isomorphic to an $n \times n$ grid (report7, Fig. 10, 2002/2003) — the construction *found by a Lisp enumerator* and drawn with CLIM (MCL), not by hand. The models are *out there*.

...but no concept can *force* them: the coincidence obstruction

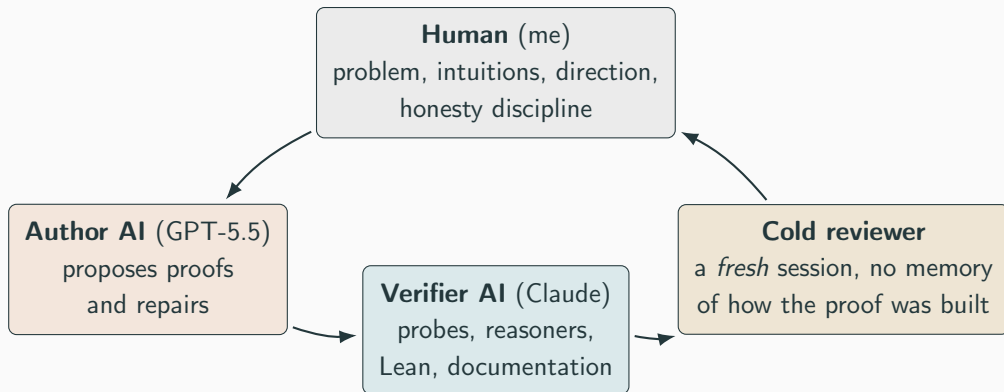


report7, Fig. 11: successors z (east-north) and z' (north-east) *should* be one corner — the table *alone* allows **six** relations, forcing neither merge nor split. You *can* force the merge (clash out the five non-EQ options, EQ remains) — but that's the trap: those markers collapse *every* corner at the next level (SAT at level 1, UNSAT at 2; RCC5 *and* RCC8). Neither gives a grid: loose $\Rightarrow$ no join, forced $\Rightarrow$ total collapse.

“Even though these infinite models do exist, we have found no way to enforce the depicted model(s) by means of a concept term.”

(report7, 2003)

The cast, and the loop



Propose → verify → **attack** → repair. Round after round, March–July 2026.

The campaign at a glance

- ~4 months, ~30 repair rounds
- 17 adversarial reviews — a defect or overclaim found in *all but two*
- 88 self-contained verification probes (each re-derives the algebra from set semantics)
- 4 Lean files, ~5,900 lines, zero sorry
- 2 working reasoners, cross-validated on 911 concepts

The single most telling number

Across *all* rounds and reviews, no accepted-but-unsatisfiable concept was ever found. Every defect was a gap in an *argument* — never a counterexample to the intended theorem.

That is why the label is *strongly supported* — and why it is still *not certified*.

Route 1: tableaux with blocking — broken by totality

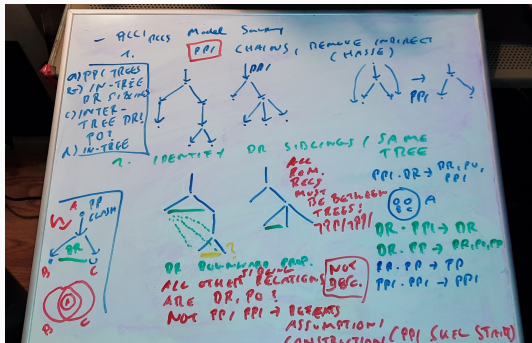
The default DL method: build a model top-down, *block* when a node repeats an ancestor's type.

- Here, composition forces edges **into already-built nodes**: from $x \text{ PP } y$, $y \text{ DR } z$ the table forces $x \text{ DR } z$.
- Diagnostic: $C_{\text{force}} = \exists \text{PP}.(\exists \text{DR}.A) \sqcap \forall \text{DR}.\neg A$ is **unsatisfiable** — any procedure that ignores forced edges wrongly accepts it.
- Dormant $\forall$'s reawaken blocked nodes; no fixed-arity local summary (types, triangles, quadruples, profiles, ...) is ever enough.

Lesson: in a total-relation logic, “local” does not exist.

What may be reused depends on the whole graph. (That is F6 in disguise.)

Route 2: the split-forest idea (a whiteboard, April 2026)



- **Present** the complete graph as a *forest of part-of trees*: split shared parts into EQ-mates, re-orient vertical edges into the trees.
- All remaining cross-edges are **horizontal**; complete them by the RCC5 *patchwork property* (a citable classical theorem).
- The one component **every** cold review found sound — the semantic foundation all later routes consume.

The human contribution: this decomposition. The models made it precise.

Route 3: tree automata — “acceptance without validity”

Package the search as a parity tree automaton (emptiness is decidable)...

- ...but the input tree **omits** the relations between fresh witnesses and the sources of $\forall$ -constraints.
- An *alternating* automaton explores in independent copies — and copies **guess the missing relations independently**.
- A cold review built the killer: two copies privately guess $\{PP\}$ and $\{DR\}$ for the same pair; $\text{comp}(PP, PP) \cap \text{comp}(PP, DR) = \{PP\} \cap \{DR\} = \emptyset$ — every copy accepts, **no model exists**.

So the automaton was **abandoned**, in favour of explicit finite certificates checked by composition tests — with a *truth lemma* instead of an acceptance condition. (This is what got formalized.)

Machine-checking: what the Lean kernel buys

- **Lean 4**: a proof assistant whose kernel type-checks *every* inference; a proof with a gap *does not compile*.
- Our discipline: Lean 4 **core only** — no libraries; every object built from first principles; tiny trusted base.
- “**zero sorry**” = no admitted gaps anywhere.

Why we moved to Lean

Most reviews broke a *prose* proof — typically at a step that was *assumed*, not proved. The kernel cannot be talked past. After the move, reviews stopped finding soundness defects and started finding only *interface* gaps — which were then closed.

What exactly is certified (zero sorry)

Result	Meaning	File
Soundness pipeline	valid certificate $\Rightarrow$ real RCC5 model	Round19Transport
Abstraction faithful	satisfiable $\Leftrightarrow$ Hintikka-realizable	Round19Transport
RCC5 normal form (both dir.)	network $\Leftrightarrow$ order + disjointness	RCC5NormalForm
Forcing reduction	F6 fails $\Leftrightarrow$ a concept forces crowds	ForcingReduction
Dovetail schema	qualitative F6 $\Rightarrow$ decidable	SemiDecidability

What is **deliberately not** in the kernel: the open completeness premise itself (F6). It is *stated*, never axiomatized — the honest boundary between what is proved and what is conjectured.

The main theorem (conditional, soundness certified)

Conditional decidability

If F6 holds — live width bounded by a computable function of the concept — then $\mathcal{ALCT}_{\text{RCC5}}$ concept satisfiability is **decidable**.

- The **soundness half** is kernel-checked end to end: certificate syntax $\rightarrow$ closure conditions $\rightarrow$ proper RCC5 frame $\rightarrow$ truth lemma $\rightarrow$ *actual model of the input concept*.
- The certificate checker is a real **executable Boolean function** — no oracle anywhere (a previous version had a hidden one; a cold review caught it; it was rebuilt).
- Everything that remains open is the **antecedent** — F6.

Unconditional win #1: a decidable fragment

Theorem ($\forall$ PO-free fragment)

Concepts with **no $\forall$ PO subformula** are decidable, with live width $\leq (1 + m)^2 2^{2^m}$,
 $m = |\text{cl}(C_0)|$.

- Still allowed: $\exists$ PO (“overlaps some. . .”), $\forall$ DR, $\exists$ DR, and **all** part-of modalities — a genuinely expressive spatial fragment.
- Why it works: PO is the *only* relation whose recurrence along a chain needs an “all-phase” safety condition; drop $\forall$ PO and the condition is vacuous.
- Found twice, independently, by two different routes — a good sign the fragment is natural, not an artifact.

Unconditional win #2: the RCC5 normal form

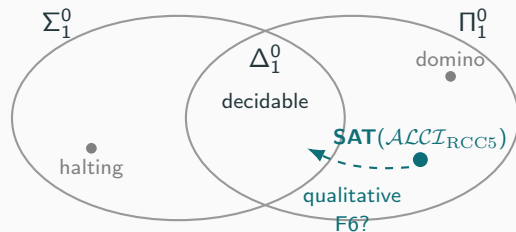
Theorem (both directions kernel-checked, arbitrary domains)

A strong-EQ RCC5 network is composition-closed *iff* it is an **ordered-disjoint structure**:

- PP is a strict partial order (a genuine part-of hierarchy);
- DR is a symmetric, **downward-closed** disjointness;
- comparable pairs never disjoint; PO is the residual.

Every RCC5 model, anywhere, *is* just: a hierarchy + a downward-inherited “keeps-apart” relation + overlap as the leftover. The “ $\Rightarrow$ ” is a 150-line kernel proof (propext only); the “ $\Leftarrow$ ” is GPT-5.6 Pro’s **canonical set representation**, also kernel-checked for arbitrary domains — the structural fact both major proof routes silently relied on, now a certified biconditional.

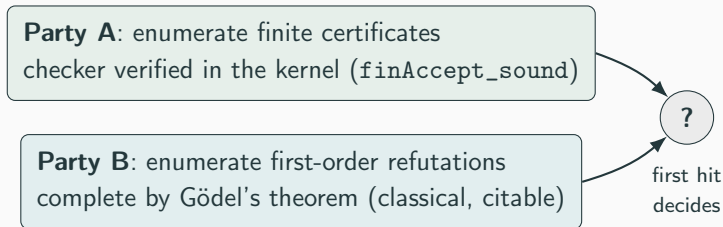
A late surprise: the problem's exact address is Π_1^0



- The whole semantics is **finitely first-order axiomatizable**; by Gödel completeness, *unsatisfiability* is enumerable.
- So SAT is **co-r.e.** (Π_1^0) — the domino problem's *ceiling* (membership; we don't claim hardness).
- Nobody in the project — or the reviews — had noticed for four months. Machine-cross-checked; the reduction is in Lean.

Consequence: half of the keystone was never needed

A Π_1^0 problem is decidable *iff* SAT is also enumerable. So run **two search parties** and wait for the first hit:



- Needed now: only **qualitative** F6 — *some* finite certificate per satisfiable concept, **no computable bound**.
- The bound the reviewers kept demanding (and that kept breaking) comes back **for free, afterwards** — a theorem of the schema.

“Isn’t this just first-order logic?” — yes, and it helps

$\mathcal{ALCI}_{\text{RCC5}}$ -SAT is FO satisfiability of a finite theory. Does it land in a **decidable fragment** of FO?

Fragment	Fate	Why
two-variable FO^2	×	composition is ternary (RCC3 <i>did</i> fall this way in 2003!)
prefix classes	×	unbounded $\forall\exists$ alternation, FMP
guarded / clique-guarded	×	totality is the one unguardable axiom
unary/guarded negation	×	finite model property; totality again
guarded + transitivity	×	undecidable outside guards

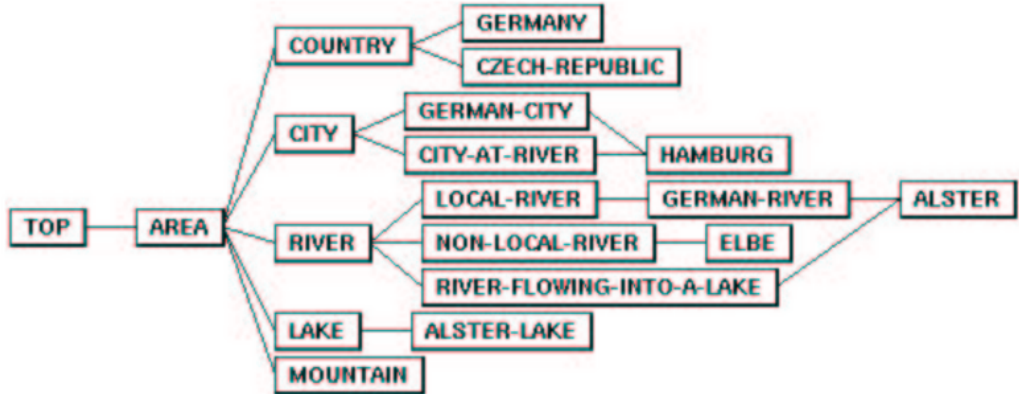
Every failure traces to the same fact: **every pair is related**, so models are maximally non-local — exactly the obstruction F6 names. The FO view rediscovers the project’s wall from classical model theory.

A working prototype reasoner (validated, not proved)

- The **cover-tree tableau** (~350 lines of Python): exact-occurrence blocking on the split-forest foundation.
- Cross-validated against an independent oracle on **911 concepts — zero mismatches**; 775/775 structured decomposition tests; never once contradicted in the whole campaign.
- Decides correctly even the known trap (the strong-EQ PO-loop) that fools type-based procedures.

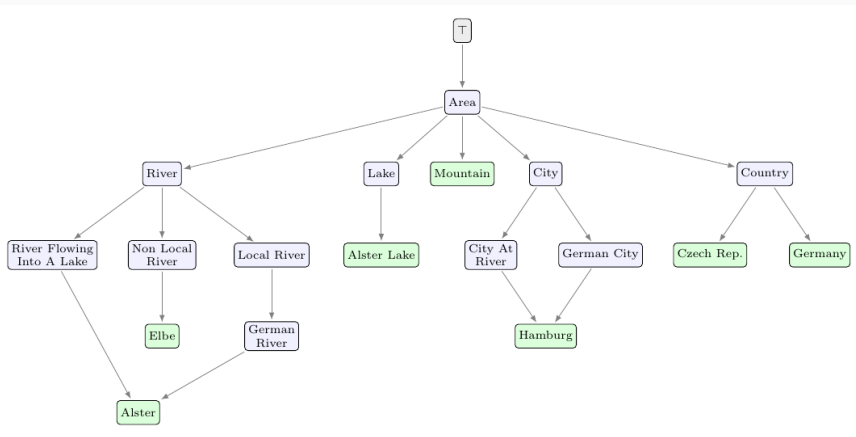
A running algorithm and a kernel-checked proof, **pointing the same way** — the strongest evidence short of a certificate.

2003: a prototype reasoner



The GIS concept taxonomy in report7 (Fig. 6), 2003 — itself *computed by a prototype reasoner* of the era, not drawn by hand.

2026, the cover-tree reasoner: all 21/21 — edge for edge



The 2026 cover-tree tableau reproduces it exactly — **two reasoners, 23 years apart, identical result**. (Green = individuals; *Hamburg* and *Alster* each have **two** parents — a DAG, not a tree.)_{31/39}

What the models did — and what they could not

Demonstrated

- months-long rigorous formal development, ~30 rounds
- genuinely useful *adversarial* review (with machine-checked witnesses)
- a 5,900-line kernel-checked Lean development
- working, validated software

Not demonstrated

- the **decisive novel idea** — F6 is uncracked
- reliable *self*-review (both models declared victory prematurely; both were caught)
- working unsupervised: the human directed every round

Neither model is self-sufficient. **What carried the project was the loop, not any single model.**

The method that worked (steal this)

1. **Cold review beats hot review.** A *fresh* session with no memory of the construction found what warm reviewers missed — every time it mattered. Make it a standard step.
2. **Don't trust — verify.** Every load-bearing claim was cross-checked mechanically: 88 probes that re-derive the algebra from set semantics, and ultimately the Lean kernel. The machine checks are the ground truth, **not the model's confidence**.
3. **Label honestly.** *Strongly supported* $\neq$ *certified*. Overclaiming happened twice; cold review caught both; the claims were withdrawn — **on the record**.

The full timestamped conversation log is public — every idea, defect, and repair traceable to its origin, human or model.

If nothing else: a message to fellow researchers

- Whatever the fate of the mathematics, this campaign is an **existence proof** of what frontier models could already do in serious formal research — *in July 2026*.
- Anyone doing serious logic or mathematics should **know** these capabilities exist. Ignoring them does not advance science; declining to use them is simply unwise.
- Saying exactly this was a stated purpose of the work. Its first outing (DL 2026) was rejected without engaging either the mathematics or the message — so it is now on arXiv, **in the open**, with the full audit trail.

The instruments have changed. The standards — verification, adversarial review, honest labels — matter more than ever.

The results, by verification level

Level	Results	What it means
Lean-certified	soundness pipeline; abstraction faithfulness; RCC5 normal form (both directions); $\forall$ PO-free lift core; $F6 \Rightarrow$ decidable; Π_1^0 dovetail	kernel-checked, with a stated axiom footprint
Theorem-level	$\forall$ PO-free decidability + bound; identity-selector characterization; $W2'$ folds into F6	proof in a work package, often with probes
Empirical	cover-tree tableau; 911-concept cross-validation; model verifier	test evidence, not a decision theorem
Open	static F6; completeness of the certificate scheme; automatic-cover property; full decidability	no proof claimed

Four levels kept rigorously distinct — conflating them is the easiest way to misread the project.

One well-posed question
can a concept force unbounded
identity-selector structure?

NO $\Rightarrow$ decidable (machinery ready, kernel-checked) YES $\Rightarrow$ the grid/undecidability route opens

- 20+ years of difficulty, compressed into one lemma — with certified machinery waiting on **both** sides of the answer.
- Everything is frozen, tagged, reproducible: paper (42pp), 4 Lean files, 88 probes, 2 reasoners, full conversation log.

Open since 2003. Smaller than ever. Still standing.

`github.com/lambdamikel/alcircc5` — thank you!

Backup: the full RCC5 composition table

comp	$DR(a, b)$	$PO(a, b)$	$EQ(a, b)$	$PPI(a, b)$	$PP(a, b)$
$DR(b, c)$	*	DR, PO, PPI	DR	DR, PO, PPI	DR
$PO(b, c)$	DR, PO, PP	*	PO	PO, PPI	DR, PO, PP
$EQ(b, c)$	DR	PO	EQ	PPI	PP
$PP(b, c)$	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
$PPI(b, c)$	DR	DR, PO, PPI	PPI	PPI	*

Row = $S(b, c)$, column = $R(a, b)$, entry = possibilities for (a, c) ; * = all five. Re-derived from finite set semantics by every probe; matches the certified Lean table cell for cell.

Backup: what a finite certificate looks like

- **Catalogue** (the finite blueprint):
 - a *core network* — the non-repeating part of the model;
 - finitely many *attachment templates* — recipes for gluing a fresh witness cluster onto an interface;
 - a *steering function* fixing the horizontal relations from old occurrences to fresh ones (the only real freedom).
- **Plan**: which template to apply where; *unfolding* it (possibly forever) produces the split-forest model.
- **Soundness**, kernel-checked: one finite check on the catalogue guarantees *every* unfolding is a legal RCC5 model.
- **Live width** = live occurrences that must coexist at one interface during unfolding — the quantity F6 must bound.

Backup: the sharpened F6 — identity-selector minors

- The naive ways to force width (shell half-graphs, prefix pumps, PO-gaps) all **collapse** to finite catalogues — machine-checked.
- The sharp obstruction: n pairs (L_i, U_i) , each kept apart by a **private** selector A_i and by nothing else — the selector incidence matrix is the $n \times n$ *identity*.
- A finite catalogue can hold only boundedly many private selectors, so: **F6 fails iff some concept forces such minors for every n .**
- Bare RCC5 networks with unbounded such structure *exist* (machine-checked) — so any F6 proof must use $\mathcal{ALCI}_{\text{RCC}}$ -*forceability*, not RCC5 consistency alone. The uniformization side-condition (W2') provably folds into F6.